Payload-Aware Two-Tier UAV Route Planning for Last-Mile Delivery

Review Report – I

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Presentation Outline

1. Introduction
2. Literature Review
3. System Design
4. Implementation and Results
5. Project Management
6. Conclusion

Background and Motivation

- Urban last-mile delivery is rapidly shifting to autonomous UAV platforms for speed, flexibility, and scalability.
- Existing routing systems treat all deliveries **identically** — a fragile medical device and a robust parcel travel the *same* route.
- Payload-specific risk, energy, and safety constraints are completely ignored by current approaches.
- Payload metadata (urgency, fragility, thermal sensitivity) encodes routing priorities, but no system exploits this information at query time.

Core Motivation

Fragmented UAV routing pipelines cannot adapt to cargo-specific constraints in real time. A two-tier architecture that bridges payload semantics to route selection addresses this gap.

Problem Statement

Problem

Existing UAV routing systems apply **static, cargo-agnostic objective weights**, resulting in **suboptimal or unsafe routes** for payloads with differing fragility, urgency, and thermal sensitivity.

- Given payload metadata $(U,\ F,\ T) \in \{1 \ldots 5\}^3$ and a weighted airspace graph $G = (V, E, W)$, find route π^* from source s to destination d that simultaneously minimises risk, distance, and energy.
- Single-objective routing cannot model conflicting objectives.
- No system dynamically translates cargo semantics into routing objective weights.
- Constraints: feasible path, energy $\leq$ battery capacity, risk $\leq$ threshold, selection latency within onboard budget.

Project Objectives

1. To build a **multi-objective A* engine (Tier 1)** that constructs a Pareto-efficient route set over a weighted airspace graph, balancing risk, distance, and energy objectives.
2. To design an **algebraic bridge** that translates payload metadata tags (urgency U, fragility F, thermal T) into normalised routing objective weights in real time.
3. To implement a **Tier 2 payload-driven selector** that scores each Pareto route using derived weights and returns the utility-maximising route π^*.
4. To evaluate the two-tier pipeline using **simulation benchmarks** against single-objective Dijkstra and A* baselines.
5. To validate the system on **real drone hardware** (Raspberry Pi 5 + Pixhawk) through outdoor flight tests.

Scope and Expected Outcomes

Included

- Payload-aware UAV route selection for single-drone delivery
- Two-tier pipeline: Pareto front construction + utility selection
- Software simulation on OpenStreetMap urban graphs (Chennai area)
- RPi 5 + Pixhawk hardware integration and outdoor flight tests
- Benchmark against Dijkstra and single-objective A*

Excluded / Constraints

- Multi-drone fleet management and coordination
- Real-time dynamic obstacle avoidance
- Commercial deployment and regulatory compliance
- Weather-based in-flight re-routing
- Deep-learning-based risk estimation

Literature Review

Author / Year	Method	Key Contribution	Limitation / Research Gap
Chokkalingam & Ganesan (2026)	Entropy-based Pareto front	Maximises route diversity via information entropy; robust across urban environments	No payload awareness; weights are static across all cargo types
Datsko et al. (2024)	Geometric Pareto spread	Uniform spread reduces clustering in the objective space	Fixed objective weights — cannot adapt dynamically to cargo semantics
Ntakolia et al. (2023)	Crowding-distance Pareto selection	Well-distributed, representative Pareto solutions via crowding distance	No semantic metadata integration at query time; selection is cargo-agnostic

Research Gap and Proposed Contribution

Identified Gaps

- No unified end-to-end pipeline for payload-aware UAV routing exists
- All cargo types receive identical routing weights
- No work maps fragility / urgency / thermal sensitivity to Pareto utility scores
- Most approaches are simulation-only; none validate on real drone hardware

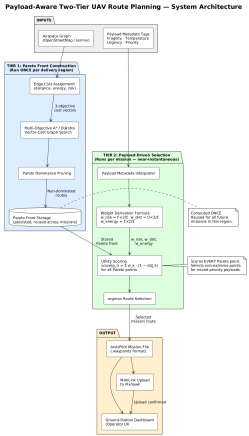
Proposed Contribution

- Two-tier pipeline: Tier 1 Pareto front + Tier 2 payload-aware selector
- Algebraic bridge: cargo metadata $\rightarrow$ normalised routing weights
- Utility-maximising selection: $\pi^* = \arg\max_{p_i \in P^*} \text{score}(p_i)$
- Full hardware validation on RPi 5 + Pixhawk drone

Existing System vs. Proposed System

Criterion	Existing Routing Systems	Proposed Two-Tier System
Routing approach	Single-objective (distance or energy only)	Multi-objective A* — risk, distance, energy simultaneously
Payload handling	Cargo-agnostic; all parcels treated identically	Payload metadata (U, F, T) drives objective weights dynamically
Route selection	Single greedy optimal path	Pareto front + utility-based selection
Adaptability	Fixed weights across all deliveries	Per-request weight derivation from cargo tags
Hardware validation	Simulation only	RPi 5 + Pixhawk outdoor flight tests

Proposed System Architecture



System Modules

1. **M1 – Data & Input Management:** Receives delivery request, parses payload metadata (U, F, T), validates inputs, and initialises the airspace graph using OpenStreetMap / OSMnx.
2. **M2 – Core Processing (Tier 1 A*):** Executes multi-objective A* search with 3-objective cost vectors $\mathbf{c}(e) = (\rho_e,\ d_e,\ \varepsilon_e)$, applies Pareto dominance pruning, and returns the Pareto front P^*.
3. **M3 – Payload Selector (Tier 2):** Applies the algebraic bridge to derive $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$; min-max normalises Pareto objectives; computes utility scores; returns π^*.
4. **M4 – UI, Hardware & Evaluation:** Provides operator dashboard (Flask / ROS 2), interfaces with RPi 5 + Pixhawk via MAVLink, and logs route metrics for benchmarking.

Methodology / Workflow

1. Literature review and problem formulation
2. Airspace graph construction (OpenStreetMap + OSMnx)
3. Multi-objective A* engine implementation (Tier 1)
4. Algebraic bridge and Pareto selector (Tier 2)
5. Software simulation and benchmarking vs. baselines
6. Hardware integration (RPi 5 + Pixhawk) and outdoor flight validation

Evaluation Metrics

- Route risk score ρ
- Route distance (km)
- Energy consumption (Wh)
- Utility score vs. baseline
- Onboard latency (ms)

Core Algorithm: Payload-Aware Two-Tier Route Planning

Algorithm Summary

1. **Input:** Graph $G = (V, E, W)$, payload tags (U, F, T), source s, destination d.
2. **Weight derivation (Algebraic Bridge):**
   $r_{\text{raw}} = F \times 2$, $d_{\text{raw}} = U \times 3$, $e_{\text{raw}} = T \times 2$; normalise $\Rightarrow w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$.
3. **Tier 1 — Pareto search:** Run multi-objective A*; accumulate cost vectors; prune dominated paths; return front P^*.
4. **Tier 2 — Scoring:** For each $p_i \in P^*$,

$$\text{score}(p_i) = w_d(1 - \tilde{d}_i) + w_\varepsilon(1 - \tilde{\varepsilon}_i) + w_\rho(1 - \tilde{\rho}_i)$$

5. **Output:** $\pi^* = \arg\max_{p_i \in P^*} \text{score}(p_i)$; generate ArduPilot mission file; upload via MAVLink.

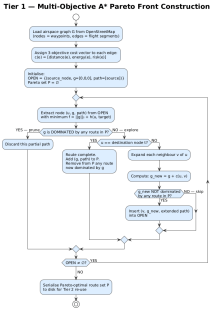
Technology Stack and Resources

Category	Selected Technology / Resource
Programming language	Python 3.11
Frameworks & libraries	ROS 2 (Humble), NetworkX, OSMnx, Flask, NumPy, SciPy
Communication protocol	MAVLink / ArduPilot `.waypoints` mission format
Map / dataset	OpenStreetMap airspace graph (Chennai urban area)
Development tools	Git, VS Code, QGIS, Mission Planner, SITL simulator, Docker
Hardware	Raspberry Pi 5 (quad-core 2.4 GHz, 8 GB), Pixhawk 6C, GPS module, ESC + motors

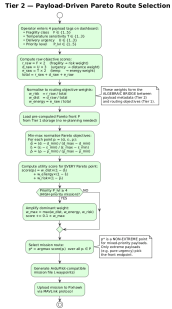
Implementation Status

Component	Status	Evidence / Remarks
Literature review & problem formulation	Completed	3 papers surveyed; gap documented
System design & architecture	Completed	Two-tier architecture defined and diagrams produced
Airspace graph construction	Completed	OSMnx graph over Chennai urban area loaded
Tier 1: Multi-Objective A* engine	In Progress	Core A* with 3-objective cost vectors implemented
Tier 2: Payload-driven selector	Planned	Algebraic bridge designed; coding next sprint
Hardware integration (RPi 5 + Pixhawk)	Planned	Hardware procured; assembly and SITL setup pending

Prototype / Implementation Diagrams



Tier 1: Multi-Objective A* Pareto Front Construction



Tier 2: Payload-Driven Pareto Route Selection

Expected Results and Performance Targets

Example: Fragile Medical Payload

Input: $F{=}5,\ U{=}4,\ T{=}3$

$r_{\text{raw}} = 10,\ d_{\text{raw}} = 12,\ e_{\text{raw}} = 6$

$w_{\text{risk}} = 0.357,\ w_{\text{dist}} = 0.429,\ w_{\text{energy}} = 0.214$

High-risk edges are **heavily penalised**. Expected: significantly safer route than shortest-path baseline.

Performance Targets

- Risk score: $\geq$20% reduction vs. Dijkstra baseline
- Pareto front size: 5–15 non-dominated routes
- Tier 2 selection latency: $<$50 ms on RPi 5
- Energy overhead: $\leq$10% above shortest-path route
- Hardware: successful outdoor delivery simulation flight

Testing and Validation Plan

Test ID	Test Scenario	Expected Result	Status
TC-01	Valid payload metadata (U, F, T) input	Correct $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ derived	Planned
TC-02	Multi-objective A* on 100-node urban graph	Pareto front P^* with zero dominated routes	Planned
TC-03	High fragility (F=5) vs. low fragility (F=1)	Significantly different route selected each time	Planned
TC-04	Disconnected graph (no feasible path)	Graceful error with clear user message	Planned
TC-05	RPi 5 MAVLink mission upload to Pixhawk	Drone accepts mission and executes flight	Planned

Validation Method

Unit tests (pytest), simulation benchmarking against Dijkstra and single-objective A*, and outdoor UAV flight tests with recorded telemetry logs.

Project Timeline (Aug 2026 – Mar 2027)

Activity	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Literature survey & formulation								
System design & architecture								
Tier 1: Multi-Objective A*								
Tier 2: Payload-Driven Selector								
Software simulation & benchmarking								
Hardware integration (RPi 5)								
Outdoor flight tests								
Documentation & report writing								

Review I: August 2026 **Final Submission:** March 2027

Team Responsibilities

Team Member	Responsibilities	Current Status
Thirumurugan RA	System architecture design, Tier 1 multi-objective A* engine, airspace graph construction (OSMnx)	In progress
Shrika SP Thota	Algebraic bridge design, Tier 2 payload selector implementation, documentation	In progress
Meghna Manimaran	Hardware procurement, RPi 5 + Pixhawk integration, testing and flight evaluation	Planned

Challenges, Risks, and Mitigation

Risk / Challenge	Possible Impact	Mitigation
Graph complexity (large OSM urban network)	Slow Pareto search	Spatial pruning, epsilon-dominance, landmark heuristics
Hardware failure (RPi / Pixhawk)	Flight tests delayed	SITL simulator validation first; spare hardware budget
Pareto front size too large	High memory and selection latency	Apply epsilon-pruning to limit front size
MAVLink integration issues	Mission upload failures	Validate protocol on SITL before real drone flights

Conclusion and Future Work

Conclusion

This project addresses the critical gap of cargo-agnostic UAV routing by proposing a two-tier payload-aware pipeline. Tier 1 builds a Pareto-efficient route set via multi-objective A*; Tier 2 selects the optimal route using an algebraic bridge that maps fragility, urgency, and thermal sensitivity into routing objective weights. System design is complete and Tier 1 implementation is underway.

Future Work

- Outdoor flight validation on a real RPi 5 + Pixhawk drone in a simulated urban delivery environment.
- Extend to multi-drone coordination with shared Pareto fronts.
- Incorporate real-time weather and dynamic obstacle data for in-flight re-routing.

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Thank You

Questions and Discussion

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